graph-theory-symbol— A set of LaTeX commands to have symbols designed for Graph Theory

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? ?

Abstract

This documentation is for package GraphTheorySymbol.

Contents

1 Usage

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix "relation".

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call. `{`⟨*commandname*⟩`}` The name of the command to create. `{`⟨*commandcode*⟩`}` The code of the command to create. This macro does nothing. It is merely an example. If this were a real macro, you would put a paragraph here describing what the macro is supposed to do, what its mandatory and optional arguments are, and so forth. Some text about an example macro called `\examplemacro`, which might have an optional argument `[`⟨*arg1*⟩`]` and mandatory one `{`⟨*arg2*⟩`}`.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value "black". `{`⟨*color*⟩`}` The color that must be used by GraphTheorySymbol afterward.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn't include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to " spoon" and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as Adj(.,.).

2 Implementation

```
⟨*package⟩
\RequirePackage{amsmath}
\RequirePackage{amssymb}
\RequirePackage{graphicx}
\RequirePackage{fdsymbol}
\RequirePackage{lua-unicode-math}
\RequirePackage{xcolor}

\def \GTS@mainColor {black}
\def \GTS@secondColor {gray}
\def \GTS@mainColorBackup {black}
\def \GTS@secondColorBackup {gray}
```

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```
\def \GTS@drawSlash {}
\def \GTS@drawUpArrow {}
\def \GTS@drawDownArrow {}

```

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call. `{`⟨*commandname*⟩`}` The name of the command to create. `{`⟨*commandcode*⟩`}` The code of the command to create.

```
\@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
  \@ifdefinable{#1}{\def#1/{#2}}
}}
```

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value "black". `{`⟨*color*⟩`}` The color that must be used by GraphTheorySymbol afterward.

```
\newcommand{\GTSsetMainColor}[1]{%
\edef\GTS@mainColorTemp{#1}%
\edef\GTS@mainColorBackup{\GTS@mainColor}%
\edef\GTS@mainColor{\GTS@mainColorTemp}%
}
```

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value "gray". `{`⟨*color*⟩`}` The color that must be used by GraphTheorySymbol afterward.

```
\newcommand{\GTSsetSecondColor}[1]{%
\edef\GTS@secondColorTemp{#1}%
\edef\GTS@secondColorBackup{\GTS@secondColor}%
\edef\GTS@secondColor{\GTS@secondColorTemp}%
}
```

`\GTSadjacency` This macro will draw an adjacency symbol.

`\GTSa`

```
\GTS@newcommand{\GTSadjacency}{%
\text{\ensuremath{%
\raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
%
\raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
$\mathcolor{\GTS@secondColor}{\circ}$%
}}%
\hspace{-0.39ex}%
\raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
$\mathcolor{\GTS@secondColor}{\bullet}$%
}}%
\hspace{-0.12ex}%
\raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
$\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
}}%
\hspace{-0.40ex}%
\raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
$\mathcolor{\GTS@secondColor}{\circ}$%
}}%
\hspace{-0.39ex}%
\raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
$\mathcolor{\GTS@secondColor}{\bullet}$%
}}%
\hspace{0.27ex}%
\if\GTS@drawUpArrow\empty\else%
\hspace{-0.65ex}%
\raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
\textcolor{\GTS@mainColor}{\textasciicircum}%
}}%
\fi%
\if\GTS@drawDownArrow\empty\else%
\hspace{-0.65ex}%
\raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
\textcolor{\GTS@mainColor}{\textasciicircum}%
}}%
\fi%
\if\GTS@drawSlash\empty\else%
```

```
\hspace{-0.65ex}%
\raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
\textcolor{black}{/}%
}}%
\fi%
%
}}%
}}%
}

\GTS@newcommand{\GTSa}{\GTSadjacency/}
```

\GTSadjacencyRelation
\GTSaR

This macro will draw an adjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSadjacency/}}
\GTS@newcommand{\GTSaR}{\GTSadjacencyRelation/}
```

\GTSunadjacency
\GTSu

This macro will draw an unadjacency symbol.

```
\GTS@newcommand{\GTSunadjacency}{%
\def\GTS@drawSlash{1}%
\GTSadjacency/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSu}{\GTSunadjacency/}
```

\GTSunadjacencyRelation
\GTSuR

This macro will draw an unadjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSunadjacency/}}
\GTS@newcommand{\GTSuR}{\GTSunadjacencyRelation/}
```

\GTSdirectedAdjacencyUp
\GTSdAU

This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyUp}{%
\def\GTS@drawUpArrow{1}%
\GTSadjacency/%
\def\GTS@drawUpArrow{}%
}

\GTS@newcommand{\GTSdAU}{\GTSdirectedAdjacencyUp/}
```

\GTSdirectedAdjacencyUp-
Relation
\GTSdAUR

This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{%
\mathrel{\GTSdirectedAdjacencyUp/}%
}
\GTS@newcommand{\GTSdAUR}{\GTSdirectedAdjacencyUpRelation/}
```

GTSnegatedDirectedAdjacencyUp
\GTSnDAU

This macro will draw a negated directed adjacency upward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{%
\def\GTS@drawSlash{1}%
\GTSdirectedAdjacencyUp/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnDAU}{\GTSnegatedDirectedAdjacencyUp/}
```

edDirectedAdjacencyUpRelation
\GTSnDAUR

This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
\mathrel{\GTSnegatedDirectedAdjacencyUp/}%
}
\GTS@newcommand{\GTSnDAUR}{\GTSnegatedDirectedAdjacencyDownRelation/}
```

\GTSdirectedAdjacencyDown
\GTSdAD

This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyDown}{%
\def\GTS@drawDownArrow{1}%
\GTSadjacency/%
\def\GTS@drawDownArrow{}%
}

\GTS@newcommand{\GTSdAD}{\GTSdirectedAdjacencyDown/}
```

directedAdjacencyDownRelation
\GTSdADR

This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{%
\mathrel{\GTSdirectedAdjacencyDown/}%
}
\GTS@newcommand{\GTSdADR}{\GTSdirectedAdjacencyDownRelation/}
```

SnegatedDirectedAdjacencyDown
\GTSnDAD

This macro will draw a negated directed adjacency downward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{%
\def\GTS@drawSlash{1}%
\GTSdirectedAdjacencyDown/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnDAD}{\GTSnegatedDirectedAdjacencyDown/}
```

DirectedAdjacencyDownRelation
\GTSnDADR

This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\mathrel{\GTSnegatedDirectedAdjacencyDown/}%
}
\GTS@newcommand{\GTSnDADR}{\GTSnegatedDirectedAdjacencyDownRelation/}
```

\GTSincidence
\GTSi

This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSincidence}{%
\text{\ensuremath{%
\raisebox{0ex}{\scalebox{1.8}[1.8]{%
%
\raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
$\mathcolor{\GTS@secondColor}{\circ}$%
}}%
\hspace{-0.39ex}%
\raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
$\mathcolor{\GTS@secondColor}{\bullet}$%
}}%
\hspace{-0.12ex}%
\raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
$\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
}}%
\hspace{-0.18ex}%
\raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
$\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
}}%
\hspace{0.2ex}%
\if\GTS@drawUpArrow\empty\else%
\hspace{-0.65ex}%
\raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
\textcolor{\GTS@secondColor}{\textasciicircum}%
}}%
\fi%
\if\GTS@drawDownArrow\empty\else%
\hspace{-0.65ex}%
\raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
```

```
\textcolor{\GTS@secondColor}{\textasciicircum}%
}}%
\fi%
\if\GTS@drawSlash\empty\else%
\hspace{-0.65ex}%
\raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
\textcolor{black}{/}%
}}%
\fi%
%
}}%
}}%
}

\GTS@newcommand{\GTSi}{\GTSincidence/}
```

`\GTSincidenceRelation` `\GTSiR` This macro will draw an incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSincidence/}}
\GTS@newcommand{\GTSiR}{\GTSincidenceRelation/}
```

`\GTSnegatedIncidence` `\GTSnI` This macro will draw a negated incidence symbol.

```
\GTS@newcommand{\GTSnegatedIncidence}{%
\def\GTS@drawSlash{1}%
\GTSincidence/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnI}{\GTSnegatedIncidence/}
```

`\GTSnegatedIncidenceRelation` `\GTSnIR` This macro will draw a negated incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedIncidenceRelation}{%
\mathrel{\GTSnegatedIncidence/}%
}
\GTS@newcommand{\GTSnIR}{\GTSnegatedIncidenceRelation/}
```

`\GTSoutIncidence` `\GTSoI` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSoutIncidence}{%
\def\GTS@drawUpArrow{1}%
\GTSincidence/%
\def\GTS@drawUpArrow{}%
}

\GTS@newcommand{\GTSoI}{\GTSoutIncidence/}
```

`\GTSoutIncidenceRelation` `\GTSoIR` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSoutIncidenceRelation}{%
\mathrel{\GTSoutIncidence/}%
}
\GTS@newcommand{\GTSoIR}{\GTSoutIncidenceRelation/}
```

`\GTSnegatedOutIncidence` `\GTSnOI` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedOutIncidence}{%
\def\GTS@drawSlash{1}%
\GTSoutIncidence/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnOI}{\GTSnegatedOutIncidence/}
```

`TSnegatedOutIncidenceRelation` `\GTSnOIR` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedOutIncidenceRelation}{%
```

```
\mathrel{\GTSnegatedOutIncidence/}%
}
\GTS@newcommand{\GTSnOIR}{\GTSnegatedOutIncidenceRelation/}
```

\GTSinIncidence
\GTSiI

This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSinIncidence}{%
\def\GTS@drawDownArrow{1}%
\GTSincidence/%
\def\GTS@drawDownArrow{}%
}

\GTS@newcommand{\GTSiI}{\GTSinIncidence/}
```

\GTSinIncidenceRelation
\GTSiIR

This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSinIncidenceRelation}{%
\mathrel{\GTSinIncidence/}%
}
\GTS@newcommand{\GTSiIR}{\GTSinIncidenceRelation/}
```

\GTSnegatedInIncidence
\GTSnII

This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedInIncidence}{%
\def\GTS@drawSlash{1}%
\GTSinIncidence/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnII}{\GTSnegatedInIncidence/}
```

GTSnegatedInIncidenceRelation
\GTSnIIR

This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedInIncidenceRelation}{%
\mathrel{\GTSnegatedInIncidence/}%
}
\GTS@newcommand{\GTSnIIR}{\GTSnegatedInIncidenceRelation/}
```

\GTSadjacencyMonochrome
\GTSaM

This macro will draw a monochrome adjacency symbol.

```
\GTS@newcommand{\GTSadjacencyMonochrome}{%
\GTSsetMainColor{black}%
\GTSsetSecondColor{black}%
\GTSadjacency/%
\GTSsetMainColor{\GTS@mainColorBackup}%
\GTSsetSecondColor{\GTS@secondColorBackup}%
}

\GTS@newcommand{\GTSaM}{\GTSadjacencyMonochrome/}
```

TSadjacencyMonochromeRelation
\GTSaMR

This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSadjacencyMonochromeRelation}{%
\mathrel{\GTSadjacencyMonochrome/}%
}
\GTS@newcommand{\GTSaMR}{\GTSadjacencyMonochromeRelation/}
```

\GTSunadjacencyMonochrome
\GTSuM

This macro will draw a monochrome unadjacency symbol.

```
\GTS@newcommand{\GTSunadjacencyMonochrome}{%
\def\GTS@drawSlash{1}%
\GTSadjacencyMonochrome/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSuM}{\GTSunadjacencyMonochrome/}
```

unadjacencyMonochromeRelation \GTSuMR This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSunadjacencyMonochromeRelation}{%
\mathrel{\GTSunadjacencyMonochrome/}%
}
\GTS@newcommand{\GTSuMR}{\GTSunadjacencyMonochromeRelation/}
```

directedAdjacencyUpMonochrome \GTSdAUM This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
\def\GTS@drawUpArrow{1}%
\GTSadjacencyMonochrome/%
\def\GTS@drawUpArrow{}%
}

\GTS@newcommand{\GTSdAUM}{\GTSdirectedAdjacencyUpMonochrome/}
```

AdjacencyUpMonochromeRelation \GTSdAUMR This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
\mathrel{\GTSdirectedAdjacencyUpMonochrome/}%
}
\GTS@newcommand{\GTSdAUMR}{\GTSdirectedAdjacencyUpMonochromeRelation/}
```

DirectedAdjacencyUpMonochrome \GTSnDAUM This macro will draw a monochrome negated directed adjacency upward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
\def\GTS@drawSlash{1}%
\GTSdirectedAdjacencyUpMonochrome/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnDAUM}{\GTSnegatedDirectedAdjacencyUpMonochrome/}
```

AdjacencyUpMonochromeRelation \GTSnDAUMR This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
\mathrel{\GTSnegatedDirectedAdjacencyUpMonochrome/}%
}
\GTS@newcommand{\GTSnDAUMR}{%
\GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
}
```

rectedAdjacencyDownMonochrome \GTSdADM This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
\def\GTS@drawDownArrow{1}%
\GTSadjacencyMonochrome/%
\def\GTS@drawDownArrow{}%
}

\GTS@newcommand{\GTSdADM}{\GTSdirectedAdjacencyDownMonochrome/}
```

jacencyDownMonochromeRelation \GTSdADMR This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
\mathrel{\GTSdirectedAdjacencyDownMonochrome/}%
}
\GTS@newcommand{\GTSdADMR}{\GTSdirectedAdjacencyDownMonochromeRelation/}
```

rectedAdjacencyDownMonochrome \GTSnDADM This macro will draw a monochrome negated directed adjacency downward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
\def\GTS@drawSlash{1}%
```

```
\GTSdirectedAdjacencyDownMonochrome/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnDADM}{\GTSnegatedDirectedAdjacencyDownMonochrome/}
```

jacencyDownMonochromeRelation
\GTSnDADMR

This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
\mathrel{\GTSnegatedDirectedAdjacencyDownMonochrome/}%
}
\GTS@newcommand{\GTSnDADMR}{%
\GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
}
```

\GTSincidenceMonochrome
\GTSiM

This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSincidenceMonochrome}{%
\GTSsetMainColor{black}%
\GTSsetSecondColor{black}%
\GTSincidence/%
\GTSsetMainColor{\GTS@mainColorBackup}%
\GTSsetSecondColor{\GTS@secondColorBackup}%
}

\GTS@newcommand{\GTSiM}{\GTSincidenceMonochrome/}
```

TSincidenceMonochromeRelation
\GTSiMR

This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSincidenceMonochromeRelation}{%
\mathrel{\GTSincidenceMonochrome/}%
}
\GTS@newcommand{\GTSiMR}{\GTSincidenceMonochromeRelation/}
```

GTSnegatedIncidenceMonochrome
\GTSnIM

This macro will draw a monochrome negated incidence symbol.

```
\GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
\def\GTS@drawSlash{1}%
\GTSincidenceMonochrome/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnIM}{\GTSnegatedIncidenceMonochrome/}
```

edIncidenceMonochromeRelation
\GTSnIMR

This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{%
\mathrel{\GTSnegatedIncidenceMonochrome/}%
}
\GTS@newcommand{\GTSnIMR}{\GTSnegatedIncidenceMonochromeRelation/}
```

\GTSoutIncidenceMonochrome
\GTSoIM

This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSoutIncidenceMonochrome}{%
\def\GTS@drawUpArrow{1}%
\GTSincidenceMonochrome/%
\def\GTS@drawUpArrow{}%
}

\GTS@newcommand{\GTSoIM}{\GTSoutIncidenceMonochrome/}
```

utIncidenceMonochromeRelation
\GTSoIMR

This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{%
\mathrel{\GTSoutIncidenceMonochrome/}%
```

```
}
\GTS@newcommand{\GTSoIMR}{\GTSoutIncidenceMonochromeRelation/}
```

negatedOutIncidenceMonochrome
\GTSnOIM

This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
\def\GTS@drawSlash{1}%
\GTSoutIncidenceMonochrome/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnOIM}{\GTSnegatedOutIncidenceMonochrome/}
```

utIncidenceMonochromeRelation
\GTSnOIMR

This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
\mathrel{\GTSnegatedOutIncidenceMonochrome/}%
}
\GTS@newcommand{\GTSnOIMR}{\GTSnegatedOutIncidenceMonochromeRelation/}
```

\GTSinIncidenceMonochrome
\GTSiIM

This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSinIncidenceMonochrome}{%
\def\GTS@drawDownArrow{1}%
\GTSincidenceMonochrome/%
\def\GTS@drawDownArrow{}%
}

\GTS@newcommand{\GTSiIM}{\GTSinIncidenceMonochrome/}
```

inIncidenceMonochromeRelation
\GTSiIMR

This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSinIncidenceMonochromeRelation}{%
\mathrel{\GTSinIncidenceMonochrome/}%
}
\GTS@newcommand{\GTSiIMR}{\GTSinIncidenceMonochromeRelation/}
```

SnegatedInIncidenceMonochrome
\GTSnIIM

This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
\def\GTS@drawSlash{1}%
\GTSinIncidenceMonochrome/%
\def\GTS@drawSlash{}%
}

\GTS@newcommand{\GTSnIIM}{\GTSnegatedInIncidenceMonochrome/}
```

InIncidenceMonochromeRelation
\GTSnIIMR

This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
\mathrel{\GTSnegatedInIncidenceMonochrome/}%
}
\GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochromeRelation/}
```

\dummyMacro

This is a dummy macro. If it did anything, we'd describe its implementation here. ”

```
\newcommand{\dummyMacro}{
```

Don't use `%` to introduce a code comment within a `macrocode` environment. Instead, you should typeset all of your comments with LaTeX—doing so gives much prettier results. For comments within a macro/environment body, just do an `\end{macrocode}`, include some commentary, and do another `\begin{macrocode}`. It's that simple.

```
}
```

```
\endinput
⟨/package⟩
```

3 Change History

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